Tribhuvan University

Institute of Science and Technology

2079

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Bachelor Level / First Year/ First Semester/ Science

Computer Science and Information Technology (PHY. 113)

(Physics)

(NEW COURSE)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
The questions are of equal value.

Section A

Long Answer Questions:

Attempt any TWO questions **(2x10=20)**

1. Explain the meaning of 'fabrication of integrated circuits'. Describe following processes involved in the fabrication of integrated circuits: epitaxial growth, oxidation, oxide removal and pattern definition, doping and interconnection of components. [10]

2. Explain the effect of external magnetic field on current carrying loops. Describe torque on a current-carrying rectangular loop of wire on a pivot rod when placed in a magnetic field. Give alternative way of increasing the torque on the coil. [10]

3. What do you mean by the wavefunction? Discuss its physical significance. Set up time-independent and time-dependent Schrodinger wave equation. What are the implications of this equation? Discuss. [10]

Section B

Short Answer Questions: **(8x5=40)**

Attempt any EIGHT questions:

4. Derive expression for electrical conductivity of semiconductor in terms of impurity ionization energy. [5]

5. Describe behavior of mobile negative charges in the Hall effect experiment. [5]

6. Set up differential equation for an oscillation of a spring using Hooke's and Newton's second law. [5]

7. What are (a) the energy, (b) the momentum, and (c) the wavelength of the photon that is emitted when a hydrogen atom undergoes a transition from the state $n = 4$ to $n = 2$? [5]

8. An oscillating block of mass 250 g takes 0.15 sec to move between the endpoints of the motion, which are 40 cm apart. Find (a) frequency and (b) amplitude of the motion, and (c) force constant of the spring. [5]

9. A potential difference of 100 V is established between the two plates one being the high potential plate (say A). A proton of charge $q = 1.6 \times 10^{-19}$ C is released from plate B, the another plate. What will be the velocity of the proton when it reaches plate A? The mass of the proton is 1.67×10^{-27} kg. [5]

10. An α-particle is emitted from a radioactive nuclei with an energy of 6.8 MeV. Calculate its wavelength and compare it with the size of the emitting nucleus that has a radius of 8×10^{-15} m. [5]

11. (a) Calculate the Fermi energy aluminum that have density 2.73 g/cm^3 and molecular weight 26.98 g/mole. (b) If the experimental value of Fermi energy(E_F) is 11.8 eV, what is the effective mass of electron in aluminum? Aluminum is trivalent. [5]

12. The output of a digital circuit (y) is given by this expression:

$$y = (\overline{AB} + \overline{B}A)(\overline{A + B} + C)$$

Where A, B and C represent inputs. Draw a circuit of above equation using OR, AND and NOT gate and hence find its truth table. [5]